



T5: SHO LOGISTICAL MODEL - DISTRIBUTION/TRANSPORT OF GOODS AND MATERIAL

T5: SHO MODEL LOGISTIKY - ROZVOZ/TRANSPORT ZBOŽÍ A MATERIALU

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Contents

1	Introduction	2
1.1	Introduction	2
1.2	Data in the model	2
2	Historical context	3
2.1	The Issue of feeding an army	3
2.2	Transportation and logistics	3
2.3	Economics	3
3	Conceptual Model	4
3.1	Diagrams	5
3.2	Diagram description	5
4	Simulation	7
4.1	Supply Chain Topology	7
4.2	Main SIMLIB Objects	7
4.3	Random Events	8
4.4	Process Classes	8
4.4.1	Convoy (derived from Process)	8
4.4.2	Town1ConvoyGenerator (Process)	8
4.4.3	Town4ConvoyGenerator (Process)	8
4.4.4	DailyConsumptionProcess (Process)	8
4.5	Configuration and Execution	9
5	Experiments	10
5.1	Army increases demand	10
5.2	Increased army demand and more effective wagons.	11
5.3	How bad weather affected the supply.	12
5.4	Faster convoys without military escort.	13
6	Conclusion	14
	Bibliography	15

Chapter 1

Introduction

1.1 Introduction

The aim of this project is to develop a discrete-event simulation model of the Roman military supply chain that transported goods and materials during Germanicus's campaigns in Germania (14–16 AD) [1]. The model focuses on the logistics system of the province *Gallia Belgica* (modern northern France and Belgium), which served as a primary rear-area hub feeding the armies operating east of the Rhine.

The central research question is: To what extent was the success of Roman offensive operations in Germania limited by the capacity and cost of overland transport, and which technological or organisational changes would have significantly reduced costs or extended the sustainable operational reach?

By combining historical data with a detailed SIMLIB/C++ simulation, the study quantifies daily consumption rates, transport capacities, monetary costs in denarii, and the impact of disruptions (weather, banditry). A reference scenario reproducing the conditions of the 15 AD campaign is first validated against known historical outcomes; subsequent experiments then evaluate proposed improvements such as higher-capacity wagons, increased use of river transport, and alternative depot placement.

1.2 Data in the model

Data used in the model and estimations gathered from multiple historical sources. Validity is checked by simulating real world scenarios and comparing them to their historical outcome. The army consumption is calculated from historical sources where by our estimate the army consumes about six hundred tones of provision a week, our simulated province needs to supply at least a third of that consumption or the provisions will decline.

Chapter 2

Historical context

2.1 The Issue of feeding an army

Germanicus's army consisted of eight legions, roughly forty thousand soldiers with the same number of auxiliaries, as well as a large number of livestock, roughly ten thousand for an army of this size. In addition, the army was accompanied by a large number of civilians who interacted with the army, often by trading. Such a large quantity of people surpassed most cities of this time and would devastate the local food supplies. For this reason, Rome distributed the burden across multiple provinces whom each provided their most produced goods and materials, these were accumulated in provinces closest to the army and distributed through a network of cities, towns, and fortified depots[3][1].

2.2 Transportation and logistics

The materials to be transported were mostly wheat, barley, livestock, animal fodder, weapons, and clothing [3]. The vast majority of these goods and materials were long lasting so they could withstand long travel times. The main means of transportation were by boats and by carriage wagons, boats could carry large amounts of cargo approximately fifty tones, wagons could carry on the average half a tone and traveled in convoys of around a hundred wagons accompanied by hundreds of mules so one convoy could carry around seventy tones on average [4]. Boats were faster more efficient and safer but reliant on close proximity to rivers and also on season so they were not always an option.

2.3 Economics

Transportation costs for getting the provisions to the army were covered by Rome itself but the goods and materials to be transported were mostly acquired as a form of provincial tax where certain portion of produce had to be given up as an army tax. [4] Due to this we decided to split the economical observations into two categories, firstly we focus on the monetary expenses covered by Rome, Rome's currency at the time was denarii [2] so that is what we will be using for monetary simulations, as for the army tax we decided to track it by number of convoys leaving the province each week, a steep increase in each would lead to unsustainable drain on the provincial or Roman economy causing decline or collapse of supply lines.

Chapter 3

Conceptual Model

As previously mentioned we are focusing on one province Gallia Belgica. The resources are accumulated in the provincial capital, modern day Reims, and from there sent out to another large city closer to the front line every few days, the next city in the supply chain will send the provisions further till they arrive at the army. In this supply chain boats would sometimes transport the provisions from Cologne to a smaller city depending on the current position of the army, for simplicity we decided to shorten the travel time and lower the travel cost between Cologne and a smaller city to account for the decrease in both that boat transportation cause.

The army consumes a certain amount of provisions each day, as long as their provisions don't reach zero the army is considered supplied and stable.

Monetary expenses are calculated each week and as long as they don't surpass a certain threshold the supply lines are considered affordable, surpassing this threshold means that the supply lines are in the long term unsustainable.

Material expenses, number of convoys dispatched per week, must not surpass a certain threshold, if that were to happen the province would become depleted leading to provincial collapse.

During each transport there is a chance for a bad weather to occur which slows the army down causing a delay, on top of that there is also a chance that the convoy will be attacked which results in portion of the cargo being lost to the thieves or bandits.

Travel time and cost between each city varies thanks to distance and road quality.

3.1 Diagrams

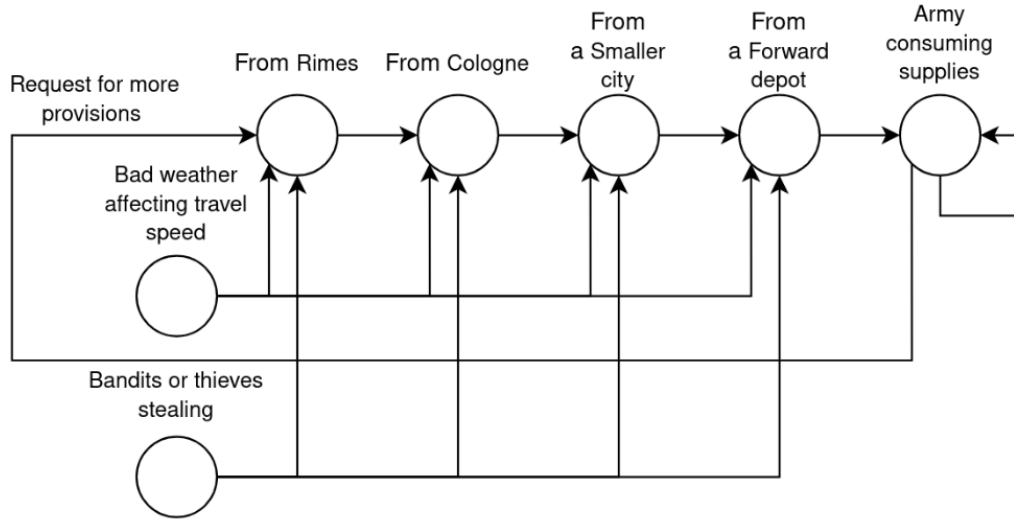


Figure 3.1: Diagram describing the whole supply chain.

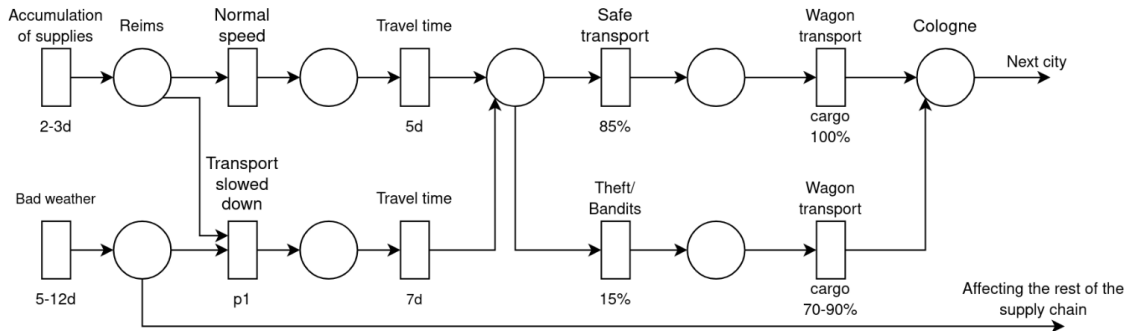


Figure 3.2: Petri Net Describing the transfer between two cities.

3.2 Diagram description

The first diagram describes the whole supply chain from start to finish, when the convoy transferring the provisions leaves a city it travels for a certain amount of time before arriving at the next city, the farther down the supply chain the provisions are the faster but more frequent transportation is, the first few convoys take days to complete while the last few between a forward depot and the army take only a few hour but consist of much smaller convoys. During this time it can be subjected to bad weather causing delays, on top of that it can be attack and/or stolen from causing effectivity to drop. At the end there is the army constantly consuming their provisions each day if it isn't properly supplied due to delays or small amount of provisions being delivered the army will run out of supplies and will need to retreat causing the campaign to fail.

The army can request for more provisions to be delivered, this will cause the province to send additional convoys, if too many convoys are sent the province will run out of resources as previously mentioned.

The second diagram Petri Net describing the transfer between two cities shows firstly that bad weather will affect the whole supply chain and can cause significant delays, historically heavy storms were known to cripple whole armies. Secondly theft and bandit attacks are something that can happen each time provisions are transported potentially depleting the whole supply, Roman army convoys always had a military escort protecting it and the provinces were generally kept safe, but decline in this safety or significant raids by enemy forces could also disrupt the whole supply.

Both of these events can happen at the same time causing significant strain on the whole supply chain.

Chapter 4

Simulation

The simulation model is implemented in C++17 using the SIMLIB discrete-event simulation library. The simulated time unit is one hour (1 day = 24 hours). The default length of one simulation run is 100 days (2400 hours), which is sufficient to observe both stable operation and collapse scenarios.

4.1 Supply Chain Topology

The logistics network is modelled as a linear chain of five nodes representing successive depots and staging points:

Town 1 (rear depot) → Town 2 → Town 3 → Town 4 → Town 5 (army position)

Travel times and fixed monetary costs per convoy on individual segments are:

Segment	Travel time [h]	Cost per convoy [denarii]
Town 1 → 2	72	2500
Town 2 → 3	24	1500
Town 3 → 4	19.2	600
Town 4 → 5	72	300

4.2 Main SIMLIB Objects

- **Facilities** ($5 \times \text{TownFacility}$) – ensure mutual exclusion during loading/unloading operations at each location.
- **Stores** ($5 \times \text{TownSupplies}$) – hold five types of supplies (grain, fodder, salted meat, wine/oil, equipment) and the number of available wagons and mules. Town 1 has practically unlimited stock; Towns 2–4 are initially empty; Town 5 represents the army’s current stockpile (initially 3000 tons).
- **Global variables** – `campaignCost`, `weeklyCostsAllRoutes[4]`, `totalRobberAttacks`, `totalSuppliesStolen`, histograms and time-series for graphs.

4.3 Random Events

- **Weather subsystem** is implemented by two alternating events:
 - **BadWeatherEvent** – sets `weatherDelayMultiplier` $\in [1.5, 2.0]$ and schedules **WeatherClearEvent** after 2–5 days.
 - **WeatherClearEvent** – resets the multiplier to 1.0 and schedules the next **BadWeatherEvent** after 10–15 days.
- **Bandit attacks** are checked after each segment is completed. Attack probabilities per segment are 50 %, 80 %, 50 %, 20 %. When an attack occurs, 20–40 % of the cargo is lost proportionally across all supply types.

4.4 Process Classes

4.4.1 Convoy (derived from Process)

Represents a single supply convoy. Its `Behavior()` method implements the following phases:

1. **Loading** – seizes the source town facility, withdraws the required amount and type of supplies, allocates wagons and mules, releases the facility.
2. **Travel** – calculates base travel time with ± 10 % uniform variation, multiplies it by the current `weatherDelayMultiplier`, adds the fixed segment cost to `campaignCost`.
3. **Attack check** – probabilistic loss of cargo (if robbers are enabled).
4. **Unloading** – seizes the destination facility, adds surviving cargo to the destination store, updates delivery statistics, releases the facility.
5. **Chain continuation** (for Towns 2–4) – immediately activates a new **Convoy** process to the next segment with the amount of cargo that the current town can provide.

4.4.2 Town1ConvoyGenerator (Process)

Periodically generates large convoys (100 wagons + 200 mules) from the rear depot (Town 1) with randomised cargo composition (40–50 t grain, 10–15 t fodder, etc.). The generation interval adapts according to the army stock level at Town 5:

- normal regime – every 56 hours (= 3 convoys/week),
- critical regime (< 500 t at Town 5) – every 42 hours (= 4 convoys/week).

4.4.3 Town4ConvoyGenerator (Process)

Generates frequent small convoys (10–20 t) from the forward depot (Town 4) to the army every 5–6 hours.

4.4.4 DailyConsumptionProcess (Process)

Runs in an infinite loop, waits exactly 24 hours, then consumes 30 tons of supplies proportionally from Town 5 storage (or everything that remains). Records daily statistics and generates all ASCII graphs at the end of the simulation.

4.5 Configuration and Execution

All important parameters are defined in `config/Config.h`. The program supports the following command-line arguments (among others):

- `-days=N` – simulation length,
- `-no-weather`, `-no-robbers` – disable the corresponding random events,
- `-seed=N` – set the pseudorandom generator seed for reproducibility.

Compilation is performed via the provided `Makefile` (`make` on Merlin/Eva or inside the supplied Docker container). Running `make run` executes the simulation with default parameters and places textual reports and ASCII graphs into the `output/` directory.

The complete source code together with the `Makefile` and the script for creating the submission archive (`make pack`) is included in the delivered archive.

Chapter 5

Experiments

In our experiments we started with higher provisions in the army at the beginning, this is to offset the supplies consumed before the first convoy arrives resulting in a sharp drop in supplies at the start of each experiment, this is an intended behavior for our simulation.

5.1 Army increases demand

In this experiment the army increased it's demand for provisions by one third causing deficit that results in higher amount of money being spent since an increasing number of convoys needs to be sent. At the same time the supply chain cannot keep up causing a complete shortage.

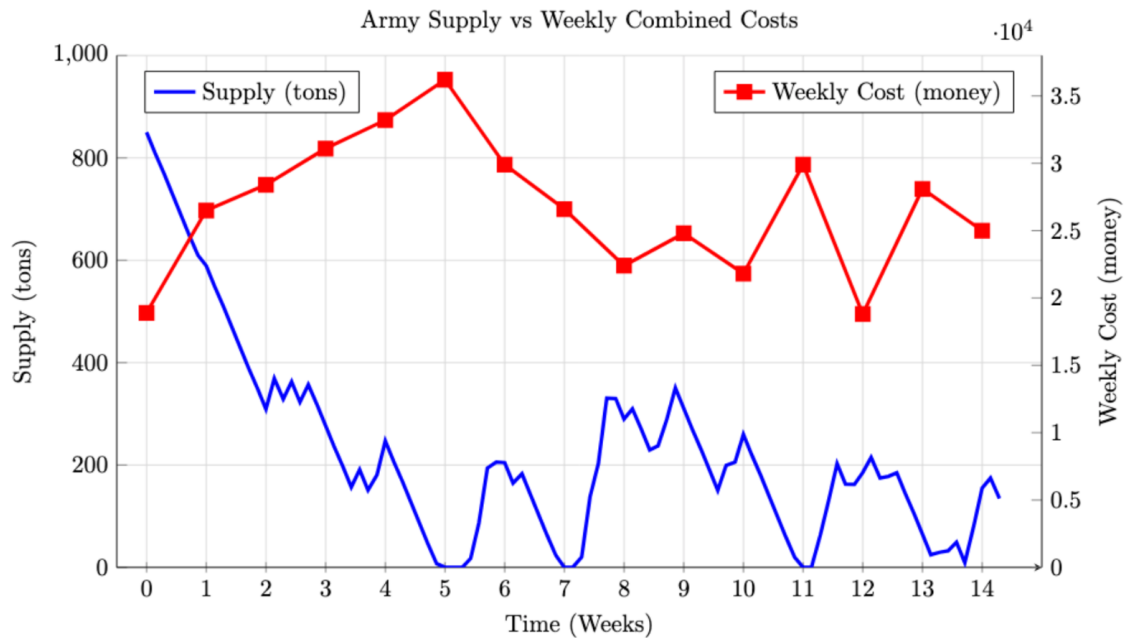


Figure 5.1: Experiment where army consumes more by 1/3.

5.2 Increased army demand and more effective wagons.

In the previous experiment the whole supply collapsed thanks to higher army demand, we wanted to know how a more effective transport could change this outcome so we changed the wagon transportation capacity to more closely match wagons found later in the medieval era.

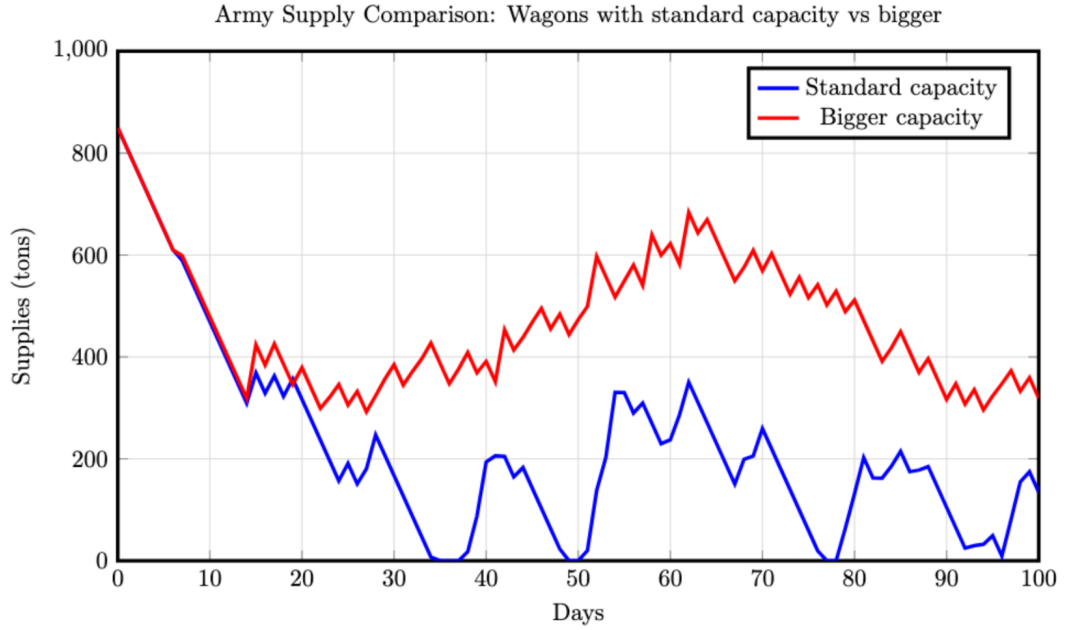


Figure 5.2: Experiment comparing Roma wagon against medieval ones.

The results show that the more advance wagons resulted in the supply chain not only sustaining the heavier load but also becoming much more stable and effective.

5.3 How bad weather affected the supply.

This experiments shows that bad weather drastically increased volatility, while normal weather resulted in stable supply, bad weather caused delays where the supply dropped before multiple convoys arrived at once causing the supply to jump up.

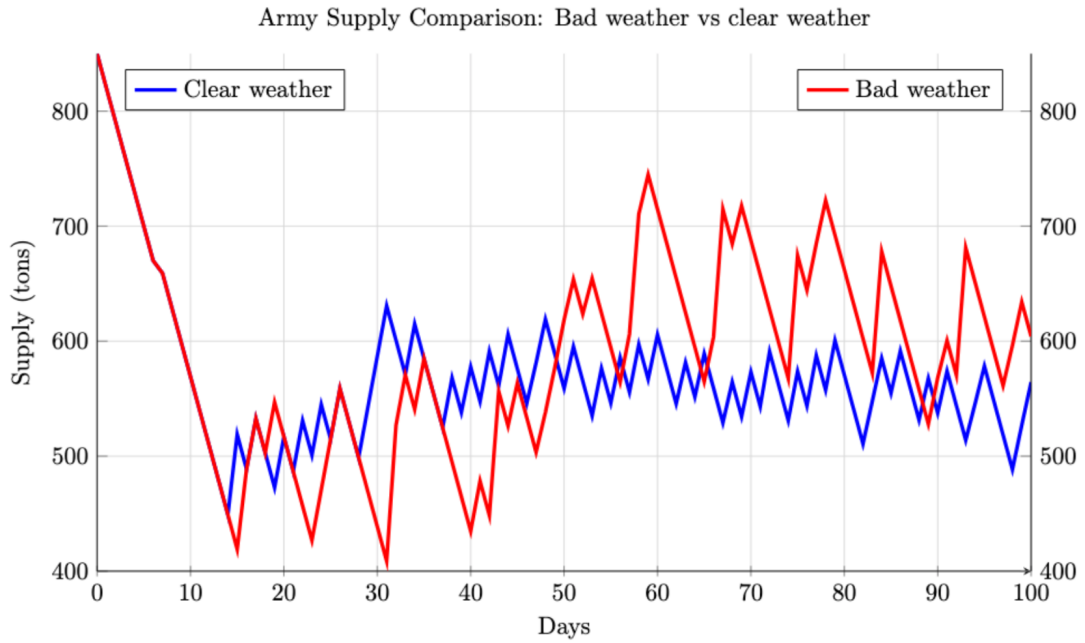


Figure 5.3: Experiment comparing good vs bad weather conditions.

This experiment shows that bad weather doesn't necessarily cause the supply chain to just slow down and halt but rather causes it to be ineffective since massive surplus would lead to a portion of the provisions being wasted while deficit would lead to shortages.

5.4 Faster convoys without military escort.

Convoys were always accompanied by a military escort to ensure their safety, this however slowed the convoy down even further. We wanted to see what would happen if the convoys didn't have additional military escort thus being significantly faster but more prone to attacks and losses. In reality this would cause moral problems thanks to frequent attacks and also be ineffective but our question is if the added speed would result in a more stable supply.

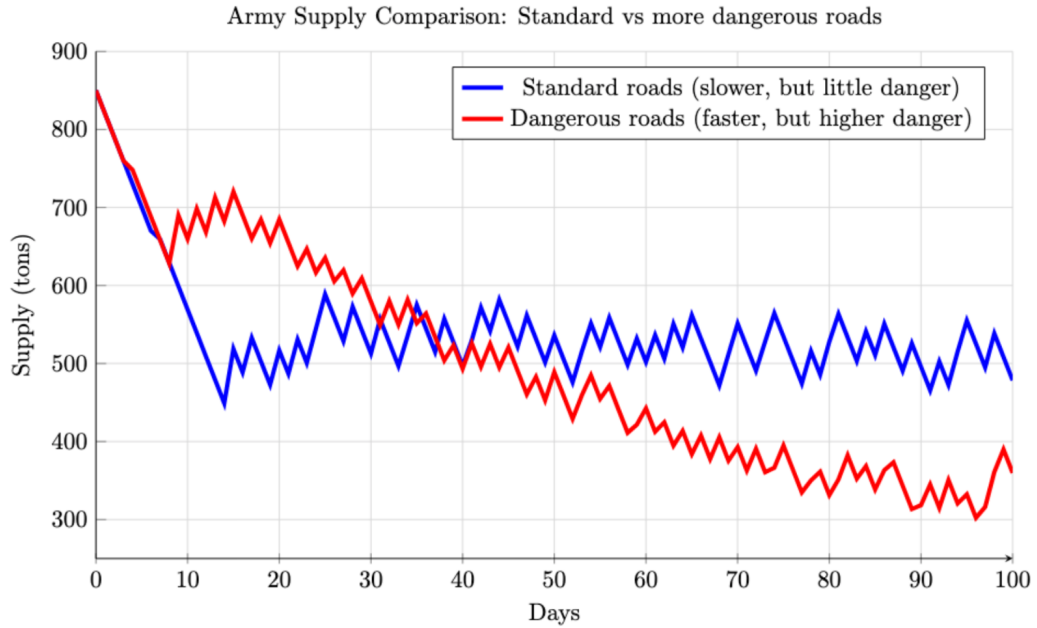


Figure 5.4: Comparison between normal and lone convoys.

The results show that while the supply did pick up faster and was more consistent, shown by the shorter frequency, the long term result was a complete decline. While not ending in complete collapse the decline would make the army more vulnerable to other factors like bad weather.

Chapter 6

Conclusion

Trough our experiments we learned that while susceptible to many disrupting factors the supply system of the Roman army was robust enough to allow the army to have stable supply line over large distances and that the Romans had many mechanisms to ensure a stable supply line. They were however very vulnerable to the effect of weather and limited by the technology of their time.

Their biggest weakness overall was inefficiency as well as dependency on the season of a year, for example monsoons could be devastating for the campaign, both of these problems could be solved by more advanced technology that could limit their reliance on weather and season as well as increase overall efficiency.

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